

Object Oriented Programming System Lab Report 2025

Lab experiment <expt_no_1>

Name: Sristee Bhindwar

Roll no.: R2142230892

Sap_ID: 500119114

B.tech CSE, Semester: 4, Batch:2

Date of experiment: 23-01-2025



School of Computer Science,
University of Petroleum and Energy Studies,
Dehradun

Introduction to Java Environment

1. Explore and understand the role of JDK, JRE and JVM.

Java Development Kit (JDK)

Role: JDK is a complete development package that includes tools for compiling, debugging, and running Java programs.

Components:

JRE (Java Runtime Environment) – To run Java applications

Compiler (javac) – Converts Java source code into bytecode

Debugger – Helps in debugging Java programs

Other tools – Javadoc, jar, etc.

Used by: Developers for writing, compiling, and debugging Java applications.

Java Runtime Environment (JRE)

Role: JRE provides the environment necessary to execute Java applications. It does not include development tools like the compiler.

Components:

JVM (Java Virtual Machine) – Runs the bytecode

Java Class Libraries – Predefined libraries required for execution

Used by: End-users who want to run Java applications but do not need to develop them.

Java Virtual Machine (JVM)

Role: JVM is responsible for executing Java bytecode and making Java platform-independent.

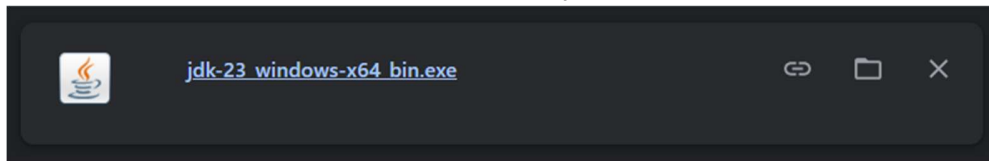
Functions:

Converts bytecode into machine code

Performs memory management (Garbage Collection)

Ensures security and portability

2. Install latest available JDK and verify Java Environment

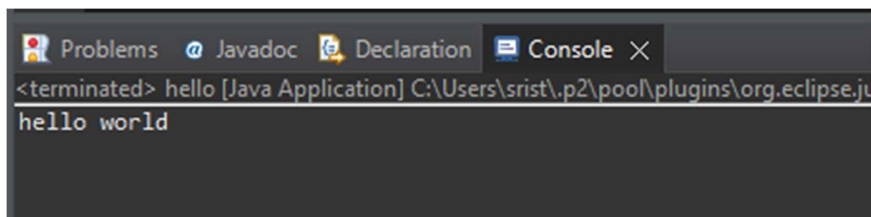


3. Create a sample Hello World Program using simple text editor and show the steps to compile and execute the program using command prompt.

CODE:

```
class Firstcode {  
    public static void main(String args[]) {  
        System.out.println("Hello World");  
    }  
}
```

OUTPUT:

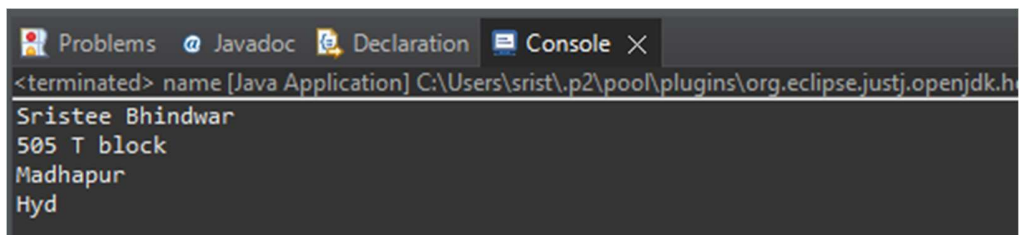


4. Display your name and complete address in different lines.

CODE:



OUTPUT:



The screenshot shows the Eclipse IDE's console window. The title bar includes tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console output is as follows:

```
<terminated> name [Java Application] C:\Users\srist\p2\pool\plugins\org.eclipse.justj.openjdk.h  
Sristee Bhindwar  
505 T block  
Madhapur  
Hyd
```